

Custom Maven Archetype – Complete Interview Guide

1) What is Custom Archetype?

A Custom Maven Archetype is a reusable project template created by developers to standardize project structure, dependencies, and configurations across multiple projects.

2) Why Do We Need Custom Archetype?

- Avoid repetitive project setup
- Maintain standard structure
- Reduce human errors
- Faster onboarding of new developers
- Ensure consistency across projects

3) Real-Time Use Case

In organizations, projects require same dependencies (Spring Boot, Logging), same folder structure, and configurations. Instead of creating manually every time, a custom archetype is used.

4) Step-by-Step Creation

Step 1: Create Base Project

Create a normal Maven project with required dependencies and structure.

```
my-project/  
  ■■■ src/  
  ■■■ pom.xml
```

Step 2: Convert Project to Archetype

```
mvn archetype:create-from-project
```

Maven converts your project into a reusable archetype.

Step 3: Generated Archetype Location

```
target/generated-sources/archetype/
```

Step 4: Install Archetype

```
cd target/generated-sources/archetype
```

```
mvn install
```

This installs archetype into local repository (~/.m2).

Step 5: Use Custom Archetype

```
mvn archetype:generate -DarchetypeGroupId=com.example -DarchetypeArtifactId=my-archetype -DarchetypeVersion=1.0-SNAPSHOT
```

5) Internal Structure of Archetype

```
archetype/  
  ■■■ src/main/resources/archetype-resources/  
  ■■■ pom.xml  
  ■■■ archetype-metadata.xml
```

Important Files

archetype-resources → Template project archetype-metadata.xml → Defines structure and files
pom.xml → Archetype configuration

6) Template Variables

```
${groupId}  
${artifactId}  
${version}  
${package}
```

These variables are replaced dynamically when project is generated.

7) Flow Diagram

Create Project → Convert → Install → Generate New Project

8) Advantages

• Reusability • Faster development • Standardization • Easy maintenance

9) Limitations

• Initial setup time required • Needs maintenance when tech changes

10) Interview Questions

Question	Answer
What is archetype?	Project template
Custom archetype?	User-defined template
Command to create?	mvn archetype:create-from-project
Where stored?	Local repository (~/.m2)

11) Final Interview Summary

Custom Maven archetype is used to create reusable project templates. It helps standardize development, reduce repetition, and improve productivity in real-time projects.